

**An Autonomous XDR Framework for Sentinel Industrial IoT
with NIST/ISO Compliance Auditing**

Project Proposal



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1 Introduction

The Industrial Internet of Things (IIoT) is rapidly changing how modern manufacturing and smart factories operate. By connecting physical machinery, sensors, and control systems to the internet, factories can achieve unprecedented levels of automation, speed, and efficiency. However, this massive web of connected devices also creates a massive target for cyber-attacks. When a hacker targets a traditional IT network, the result is usually stolen data. But when a hacker targets an Industrial IoT environment, the results can be catastrophic—leading to physical machine damage, production shutdowns, or severe safety hazards for workers.

To protect these critical environments, most organizations currently rely on traditional Intrusion Detection Systems (IDS). Unfortunately, these older systems have a major flaw: they only monitor the "surface level" of the network [1]. They look at data packets traveling back and forth, but they are completely blind to what is happening inside the actual machines and sensors [2]. Furthermore, these systems are entirely passive. When they detect a threat, they simply send an alert to a human operator and wait for someone to fix the problem. In a high-speed factory where attacks happen in milliseconds, waiting for a human to read an alert and manually block an attack is far too slow [3].

To solve these critical issues, this project introduces Sentinel-IoT, an autonomous Extended Detection and Response (XDR) framework [1]. Sentinel-IoT is designed to act as a self-driving security guard for smart factories. It uses a powerful artificial intelligence model (CNN-BiLSTM-Transformer) to analyze data from the network, the computer logs, and the physical sensors all at the same time, allowing it to catch hidden and "Zero-Day" attacks [4, 5]. More importantly, when Sentinel-IoT finds a threat, it uses Wazuh [6] to automatically block the hacker and kill the malicious process in less than a second, without needing any human help [7]. Finally, it records every action on a custom Governance, Risk, and Compliance (GRC) dashboard to prove the factory is legally compliant with international security rules like NIST [8] and ISO 27001 [9].

2 Literature Review

Organizations have historically relied on traditional Intrusion Detection Systems (IDS) utilizing signature-based detection rules for securing critical infrastructure. While tools like Snort and Suricata provide foundational network monitoring, they suffer from a critical limitation: they only analyze surface-level traffic and cannot detect sophisticated, unknown zero-day attacks. Furthermore, as Davis notes, these legacy systems are fundamentally passive; they generate alerts but lack the capability to execute active cyber defense or automated mitigation, leaving high-speed industrial environments vulnerable during the crucial window between detection and manual human response [1].

Researchers introduced machine learning techniques to identify complex anomalies and address the shortcomings of signature-based methods. Early implementations of deep learning IDS successfully utilized standard neural networks to detect previously unseen threats based on behavioral deviations rather than static signatures [10]. However, these models introduced new challenges. They function primarily as "black boxes" lacking explainability, which hinders trust in critical industrial settings. Moreover, they remained detection-only frameworks, failing to integrate active enforcement mechanisms or correlate network data with internal host telemetry.

Recent advancements have significantly improved detection accuracy within the Industrial Internet of Things (IIoT). For instance, Alashjaee proposed a highly accurate Attention-CNN-LSTM architecture capable of identifying complex threats with sub-35ms latency [2]. Despite this high performance, the study acknowledges a crucial gap: the framework lacks an automated active mitigation system and relies on limited feature attention weights rather than comprehensive explainability. Sentinel-IoT directly addresses this research gap by integrating a superior hybrid AI engine (CNN-BiLSTM-Transformer) with an autonomous Extended Detection and Response (XDR) platform [6]. This transition from passive monitoring to sub-second, explainable remediation—coupled with automated compliance auditing—forms the core contribution of the proposed framework.

Table 1 summarizes this literature, comparing traditional IDS, standalone XDR platforms, and recent deep learning frameworks against the proposed Sentinel-IoT architecture.

Table 1: Comparative Analysis of Industrial Security Solutions

Study / Solution	Approach	Detection Capability	Automated Remediation	Explainability & GRC	Key Limitation
Traditional IDS (Snort/Suricata) [1]	Signature-based detection	Known threats only	No (Passive alerting)	No	Cannot detect zero-day attacks; static rule-based system.
Deep Learning IDS [10]	Standard Neural Networks	Yes (ML-based anomaly detection)	No	No (Black box model)	Detection-only approach; no active enforcement mechanism.
Standalone XDR (Wazuh Out-of-the-box) [6]	Rule-based endpoint telemetry	Limited (Depends on static rules)	Yes	Limited (Logs only)	Struggles with complex, hidden IoT anomalies without advanced AI.
Base Paper [2]	Attention-CNN-LSTM	Yes (High accuracy on Bot-IoT/NSL-KDD)	No (Detection-only; sub-35ms latency)	Limited (Feature attention weights only)	Lacks automated active mitigation; recommends testing on newer datasets (ToN_IoT).
Proposed Sentinel-IoT System	Hybrid AI (CNN-BiLSTM-Trans.) + XDR	Yes (Zero-Day capability)	Yes (Sub-second response)	Yes (SHAP graphs + NIST/ISO)	Computationally intensive; evaluated on simulated data (ToN_IoT).

As demonstrated in Table 1, there is a clear evolutionary progression from passive, signature-based monitoring to advanced AI-driven frameworks within industrial security. While recent deep learning models and standalone XDR platforms address isolated challenges—such as zero-day detection or basic rule-based remediation—they fail to provide a cohesive, transparent defense mechanism. The proposed Sentinel-IoT system uniquely bridges these remaining gaps by combining a high-speed hybrid AI engine with autonomous, sub-second remediation and comprehensive, mathematically backed GRC auditing capabilities.

3 Project Objectives

To address the critical security and compliance gaps in modern Industrial IoT environments, this project proposes the development of Sentinel-IoT. The specific objectives required to build this autonomous XDR framework are as follows:

- **To develop a Hybrid AI Detection Engine:** To design and implement a CNN-BiLSTM-

Transformer deep learning model capable of performing both early attack prediction and zero-day anomaly detection within high-speed Industrial IoT networks [4, 11].

- **To eliminate "Endpoint Blindness":** To engineer a multi-layered data ingestion pipeline that simultaneously correlates network traffic, host operating system logs, and physical sensor telemetry using the ToN_IoT dataset [12].
- **To engineer an Autonomous Active Response System:** To integrate an Extended Detection and Response (XDR) mechanism (via Wazuh) that instantaneously neutralizes verified threats—such as blocking malicious IP addresses and killing compromised process IDs (PIDs)—without requiring human intervention [7, 3].
- **To establish a real-time GRC Auditing Interface:** To build a web-based dashboard that automatically maps detected security incidents to international regulatory frameworks (NIST SP 800-53 and ISO 27001) to generate automated compliance reports [13, 9].
- **To integrate Explainable AI (XAI):** To implement SHAP (SHapley Additive exPlanations) to resolve the AI "Black Box" problem, providing security analysts with visual, mathematically backed proofs for every automated defensive action taken by the system [14, 15].

4 Problem Description

Traditional Industrial IoT security systems, including recent deep learning models, rely heavily on passive network traffic analysis [2]. This limited scope creates a critical "endpoint blindness," rendering them ineffective against sophisticated Zero-Day attacks and internal threats that bypass standard firewalls [1]. Without correlating network data with the internal logs of the host machines and sensors, these systems cannot see the full picture of an attack [16].

Furthermore, existing research is largely reactive, lacking both early attack detection and autonomous active response capabilities [3]. Most systems wait for an attack sequence to fully materialize before simply alerting human operators, failing to identify and stop precursor activities [17]. This creates a dangerous latency between detection and remediation. In high-speed industrial environments, this delay allows malicious processes to execute catastrophic hardware failures or data theft before manual human intervention can occur [5].

Finally, the deployment of "Black Box" AI in critical infrastructure hinders trust and regulatory alignment [14]. Current solutions fail to explain their security decisions or map technical alerts to strict governance frameworks like NIST SP 800-53 and ISO 27001 [15]. This prevents the automated generation of audit trails, leaving factory managers unable to provide the legal proof of security required for compliance [13].

5 Methodology

The methodology for the Sentinel-IoT project follows a structured, modular approach designed to bridge the gap between advanced deep learning and real-time industrial security. This section outlines the architectural framework and the specific developmental phases required to achieve a resilient and autonomous security system.

5.1 Approach

The development of the Sentinel-IoT framework relies on a modular, multi-layered software engineering approach designed to bridge the gap between threat detection and automated remediation. The system integrates a custom deep learning engine (CNN-BiLSTM-Transformer) with the open-source Wazuh Extended Detection and Response (XDR) platform. This architecture ensures that network traffic, host operating system logs, and physical sensor telemetry from the ToN_IoT dataset are processed simultaneously [12]. By combining artificial intelligence with active endpoint response, the framework moves beyond passive alerting to trigger real-time, autonomous defensive actions [3].

The overall data flow begins with the ingestion of the ToN_IoT dataset and live telemetry streams through the MQTT protocol [12]. The AI engine preprocesses and transforms the data into structured sequences before performing anomaly detection [11]. Once a threat score exceeds a defined threshold, the AI engine sends a trigger signal via REST API to the Wazuh XDR Manager [7], which executes predefined defensive scripts and logs compliance mapping records in the central dashboard [13].

5.2 System Architecture

The high-level architecture of the Sentinel-IoT framework is illustrated in Figure 1. The diagram depicts the multi-layered data ingestion pipeline [16], the core hybrid AI detection engine (CNN-BiLSTM-Transformer) [4], and the automated remediation flow triggered via the Wazuh XDR platform [1].

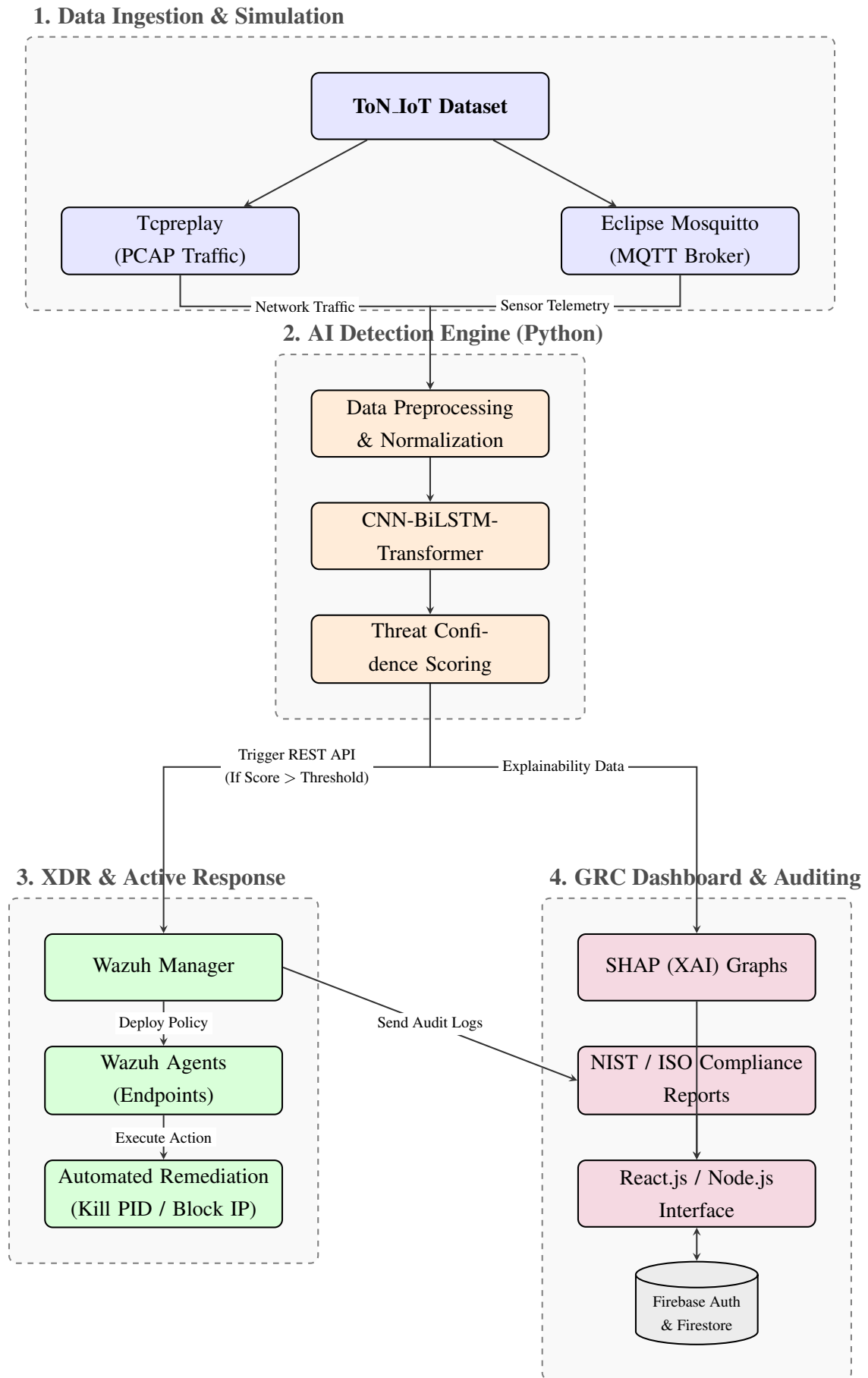


Figure 1: High-Level System Architecture of the Sentinel-IoT Framework

Figure 1 illustrates the four-stage data pipeline of the Sentinel-IoT framework, progressing from raw telemetry ingestion to AI-driven anomaly detection [16]. Upon identifying a threat, the AI engine simultaneously triggers autonomous remediation via the Wazuh XDR platform and routes explainability data to the GRC dashboard for compliance auditing [1, 15]. While the Wazuh XDR platform acts as the primary data store for security logs, Firebase is integrated with the Node.js backend to securely manage the web application’s user authentication and role-based access control.

5.3 Development Model

The project utilizes the **Iterative** development model for its development lifecycle. A traditional Waterfall model was not selected because the performance of machine learning models cannot be fully predicted during the initial planning stages and inherently requires continuous retraining and validation cycles [18].

Under the Iterative model, the Sentinel-IoT framework will be developed and improved through successive refinement cycles:

- **Initial Iterations (AI Core):** The first cycles will focus on developing a baseline AI detection engine. The model will undergo repeated training and tuning iterations to adjust sequence lengths and hyperparameters until optimal zero-day threat detection is achieved [4, 5].
- **Subsequent Iterations (Integration & Tuning):** Later iterations will expand the system by integrating the Wazuh active response mechanisms and the React.js GRC auditing dashboard. Within these cycles, defense scripts and API communications will be continuously tested and refined until the framework reliably meets the sub-second threat termination requirement [3].

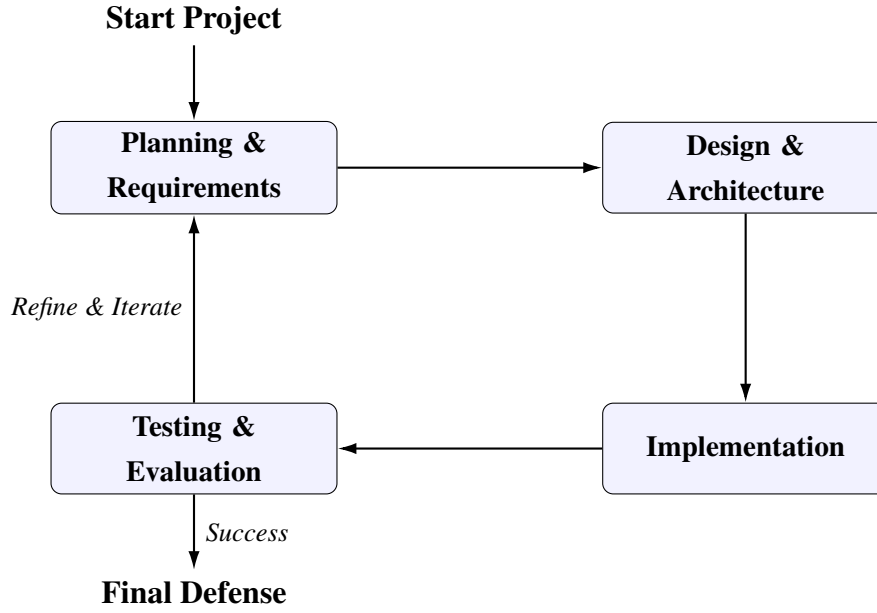


Figure 2: Iterative Software Development Process Model for Sentinel-IoT

Figure 2 illustrates the iterative lifecycle of software development tailored for the Sentinel-IoT framework. The process flows through continuous cycles of planning, architectural design, core implementation of the AI and Wazuh components, and rigorous system evaluation. If the framework fails to meet strict sub-second latency or threat detection benchmarks during testing, the cycle iterates back to the requirements phase for continuous refinement until the project is successful and ready for the final defense.

5.4 User Roles

The system operates within a Security Operations Center (SOC) environment, utilizing a combination of human operators and automated system actors. To establish clear boundaries for the Use Case diagram, the following roles are defined:

Primary Actors (Human Roles)

- **System Administrator (IT Engineer):** Exercises full control over the infrastructure, including capturing data streams, approving security policies, and deploying Wazuh agents across factory virtual machines [6].
- **Security Analyst (Engineer):** Responsible for daily monitoring of real-time alerts, reviewing Explainable AI (SHAP) visualizations, suggesting new detection rules, and simulating attack tests to validate system resilience [14].
- **Compliance Officer (Auditor):** A non-technical role focused on governance, responsible for accessing the GRC dashboard to evaluate security posture and exporting automated NIST/ISO audit reports [15].

Secondary Actors (System Roles)

- **Sentinel AI Engine:** The autonomous "brain" that ingests data, performs multi-layered anomaly detection, and generates threat confidence scores [4].
- **Wazuh XDR Manager:** The automated "muscle" that receives trigger signals to execute active response scripts (e.g., terminating malicious processes) and logs compliance mapping [3, 7].

5.5 System Interaction Model

To maintain operational safety in high-speed industrial settings, the framework utilizes strict threshold logic and fail-safe mechanisms. If the AI engine calculates a threat confidence score above a critical threshold (e.g., > 0.85), Wazuh autonomously executes remediation [1]. However, if the confidence score falls into a marginal range (e.g., between 0.60 and 0.84), the system generates a high-priority warning alert without executing automated remediation. This fail-safe allows for human verification, ensuring that critical factory processes are not inadvertently interrupted by false positives [18].

5.6 Model Evaluation and Performance Metrics

The hybrid AI model will be rigorously evaluated against baseline architectures (such as standard CNN-LSTM models) using the following metrics:

- **Classification Metrics:** Model predictive performance will be measured using Accuracy, Precision, Sensitivity (Recall), Specificity, F-1 Score, and ROC-AUC [5].
- **Latency Benchmarks:** Operational speed will be evaluated by measuring the Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR), ensuring total latency strictly adheres to the sub-second requirement [2].

6 Project Scope

The scope of the Sentinel-IoT framework is focused on the design, development, and validation of an autonomous Extended Detection and Response (XDR) security system tailored for Industrial IoT (IIoT) environments.

Assumptions

The evolution and evaluation of this system are based on the assumption that it will operate within a distributed, virtualized environment. Furthermore, we assume the availability of high-fidelity, pre-captured datasets that accurately simulate the combined network, operating system, and sensor behaviors of a real-world smart factory.

In-Scope (Included)

The following design, development, and research aspects are included in this project:

- **Multi-layered Data Correlation:** The system will analyze and correlate telemetry across three distinct layers: network traffic, operating system behaviors, and simulated physical sensor data [12, 16].
- **Hybrid Anomaly Detection:** Development of an AI-driven detection engine capable of identifying both known threats and Zero-Day anomalies (such as DDoS, ransomware, and data injection) specific to industrial operations [11, 5, 17].
- **Autonomous Active Response:** Engineering an automated remediation mechanism that can independently execute defensive actions, such as terminating malicious processes or blocking unauthorized connections, upon threat verification [3, 7].
- **Automated Compliance Auditing:** Designing a governance reporting dashboard that maps detected security incidents directly to standard regulatory frameworks (NIST SP 800-53 and ISO 27001) for automated auditing [8, 13, 15].

Out-of-Scope (Not Included)

To maintain the feasibility of the project timeline, the following aspects will not be considered:

- **Physical Hardware Deployment:** The project will not involve the physical installation, testing, or integration of actual factory machinery, Programmable Logic Controllers (PLCs), or live IoT sensors [2].
- **Custom Communication Protocols:** The research will not focus on the development or optimization of the underlying physical communication media or low-level network transmission protocols.
- **Physical Security Measures:** The framework focuses exclusively on cyber-threats and will not address or detect physical security breaches (e.g., unauthorized human access to server rooms or factory floors).

7 Feasibility Study

A comprehensive feasibility study was conducted to ensure that the Sentinel-IoT project is both technically viable and practically implementable within the constraints of a smart factory environment. The following analysis evaluates the operational requirements and potential challenges associated with deploying a hybrid AI-XDR system.

7.1 Risks Involved

While the project is technically viable, the following risks must be managed to ensure success:

- **Data Imbalance Risk:** The ToN_IoT dataset may contain significantly more "normal" traffic than "attack" traffic, which can bias the AI model and lead to high false-negative rates [12, 18].
- **Integration Latency:** There is a risk that the communication between the Python AI engine and the Wazuh XDR platform via REST APIs may exceed the sub-second response requirement due to network overhead [7].
- **Computational Intensity:** The CNN-BiLSTM-Transformer architecture is resource-heavy; there is a risk that the local hardware may struggle with real-time inference during the testing phase [4].
- **Overfitting:** The model may perform exceptionally well on the ToN_IoT dataset but fail to generalize to real-world industrial traffic or novel zero-day threats [18].

7.2 Resource Requirements

The project relies on a combination of existing data, open-source software, and standard hardware:

- **Hardware:** A workstation with at least 16GB RAM and a dedicated GPU (e.g., NVIDIA RTX series) to support the training and iteration of the deep learning model [4].
- **Software (Backend):** Python 3.x environment with deep learning libraries (PyTorch or TensorFlow) for the AI engine development [14].
- **Security Platform:** Wazuh Manager and Agents installed on Ubuntu Linux virtual machines to handle endpoint monitoring and active response [6].
- **Dataset:** The ToN_IoT dataset, which provides the necessary network, OS, and sensor telemetry for training and evaluation [12].
- **Frontend:** React.js framework for developing the GRC Auditing and XAI dashboard [9, 15].
- **Database & Authentication:** Firebase (Firebase Auth and Cloud Firestore) to securely manage user authentication, session states, and Role-Based Access Control (RBAC) for the web dashboard actors.

8 Solution Application Areas

- **Smart Manufacturing and Industry 4.0:** Sentinel-IoT is primarily designed for high-speed industrial environments like smart factories. It protects automated assembly lines and robotic systems from cyber-physical attacks that could cause equipment damage or costly production downtime, ensuring the safety of both machinery and personnel [5, 13].
- **Critical Infrastructure and Utilities:** The framework is also applicable to essential sectors such as power grids, water treatment plants, and energy facilities. By providing autonomous threat neutralization and NIST-compliant auditing, it helps secure vital national infrastructure against sophisticated cyber threats while meeting strict regulatory requirements [8, 9].

8.1 Tools and Technologies

The software stack for the Sentinel-IoT framework is categorized by system components to ensure a seamless transition from real-time data ingestion to automated response.

Table 2: Sentinel-IoT Development and Deployment Software Stack

Component	Tool / Technology	Primary Function and Purpose
Deep Learning & AI Backend	Python (3.11.9)	Core programming language for the anomaly detection engine.
	PyTorch / TensorFlow	Frameworks used to build and train the hybrid CNN-BiLSTM-Transformer architecture [4, 11].
	SHAP	Explainable AI (XAI) tool for generating visual interpretations of model decisions [14, 15].
	Pandas & Scikit-learn	Libraries for data preprocessing, feature normalization, and model evaluation.
Endpoint Security & Active Response	Wazuh XDR Platform	Agent-based endpoint monitoring, automated remediation execution, and NIST/ISO mapping [6, 1].
	Ubuntu Linux (VMs)	Primary OS environment for hosting the Wazuh Manager, AI backend, and factory endpoints.
Data Ingestion & Communication	Suricata	A high-performance NIDS used for real-time network packet capture and protocol analysis.
	Tcpdump	Command-line utility for raw PCAP capture to support forensic verification and latency benchmarking.
	Eclipse Mosquitto	MQTT broker used to handle lightweight IoT sensor telemetry from simulated devices [12].
	Tcpreplay	Utility used to replay dataset PCAP files from Kali Linux to simulate live network traffic [12].
	REST APIs / Web-hooks	Communication protocols bridging the Python AI engine with the Wazuh Manager for response execution [7].
Frontend Auditing Dashboard	React.js	Frontend library used to build the web-based GUI for SOC analysts.
	Node.js	Backend runtime environment supporting the dashboard and GRC auditing interface [9, 15].
	Firebase (Auth & Firestore)	Serverless database and authentication platform used to securely manage user credentials, session states, and Role-Based Access Control (RBAC) for the dashboard.

As demonstrated in Table 2, the Sentinel-IoT framework utilizes a multi-layered software stack to bridge the gap between complex AI analysis and real-time industrial response. This architecture ensures that raw network and sensor data are not only captured and classified with

high precision but are also translated into immediate, automated mitigation actions and transparent compliance audits. By integrating specialized tools for ingestion, deep learning, active response, and dashboard user management, the system provides a comprehensive, end-to-end security solution for modern smart factories.

8.2 Dataset Selection

The primary dataset selected for training and validating the AI model is the **ToN_IoT dataset** [12]. Unlike traditional intrusion detection datasets (e.g., NSL-KDD) that focus solely on standard IT network traffic, ToN_IoT provides a heterogeneous collection of data critical for modern industrial environments [16]. It includes network traffic logs (PCAP), operating system logs from Windows and Linux endpoints, and physical telemetry data from IoT sensors [12]. Furthermore, it contains modern cyber-physical attack vectors, including ransomware, DDoS, and data injection [11]. Utilizing ToN_IoT directly addresses the future work recommendations identified in the primary base study [2], upgrading the evaluation from older benchmarks to a modern IIoT standard.

8.3 Dataset Overview

The following table outlines the structural properties and specific role of the ToN_IoT dataset within the proposed framework.

Table 3: Dataset Characteristics and Utilization

Dataset Name	Devices/ Size	Traffic Type	Attack Types	No. of Features	Key Features Included	Purpose in Project
ToN_IoT [12]	Heterogeneous IoT sensors, Linux/ Windows VMs (Large-scale)	Network (PCAP), OS Logs, Sensor Telemetry (MQTT/ Modbus)	DDoS, Ransomware, Injection, Backdoor, XSS, Password, etc.	43+ (Network subset)	Protocol, IP/Port, Byte counts, MQTT topics, system process metrics	Training the AI engine and simulating live industrial sensor streams via Mosquitto.

9 Expertise of the Team Members

The successful execution of the Sentinel-IoT project relies on the diverse and complementary skills of our team. The members bring a solid foundation in computer science and software engineering, with specific strengths divided across three core areas:

- **Machine Learning and Data Analysis:** The team has practical experience in processing large datasets and training artificial intelligence models using libraries such as TensorFlow and PyTorch. This expertise is crucial for teaching the system how to distinguish between normal factory operations and potential cyber threats.

- **Cybersecurity and Systems Administration:** The team possesses a strong understanding of network security principles and endpoint protection. This includes the ability to configure virtual machine environments and integrate security platforms that can actively block detected attacks.
- **Software Development and Integration:** The team is skilled in web development (React.js) and system integration. This ensures that the complex AI background processes can be connected to a clean, user-friendly dashboard that a security analyst or auditor can easily understand and operate.

10 Milestones

The development of the Sentinel-IoT framework is structured around key milestones, aligning with the Hybrid Agile methodology. Each major milestone represents the successful completion and integration of a core functional block across two main project iterations: FYP 1 and FYP 2.

10.1 Work Breakdown Structure (WBS)

The following Work Breakdown Structure divides the project lifecycle into manageable phases and tasks, providing an estimated duration for each step to ensure timely completion before the final defense.

Table 4: Work Breakdown Structure (WBS)

WBS ID	Task Description	Duration (Weeks)
Iteration 1: Final Year Project 1 (FYP 1)		
1.0	Project Initiation & Planning	3 Weeks
1.1	Literature Review & Requirement Analysis [2]	2 Weeks
1.2	Project Proposal Formulation & Approval	1 Week
2.0	Data & AI Engine	6 Weeks
2.1	ToN_IoT Dataset Acquisition & Preprocessing [12]	2 Weeks
2.2	CNN-BiLSTM-Transformer Architecture Design [4, 11]	2 Weeks
2.3	Model Training, Testing, and Tuning [5]	2 Weeks
3.0	XDR Integration	5 Weeks
3.1	Ubuntu Linux & Wazuh Environment Setup [6]	2 Weeks
3.2	REST API Development for AI-Wazuh Communication [7]	2 Weeks
3.3	Automated Remediation Scripting (Process Kill/Block) [3]	1 Week
Iteration 2: Final Year Project 2 (FYP 2)		
4.0	Dashboard & GRC	5 Weeks
4.1	React.js Frontend Interface Development	2 Weeks
4.2	SHAP (Explainable AI) Visualizations Integration [14]	1.5 Weeks
4.3	NIST/ISO Compliance Auditing Module Development [9, 15]	1.5 Weeks
5.0	System Evaluation	3 Weeks
5.1	Live Stream Simulation Setup via Eclipse Mosquitto [12]	1 Week
5.2	Full System Latency & Accuracy Benchmarking [2, 13]	2 Weeks
6.0	Documentation & Final Submission	3 Weeks
6.1	Final Report Writing & Thesis Formatting	2 Weeks
6.2	Final Defense Presentation & Demo Preparation	1 Week
Estimated Total Project Duration:		25 Weeks

Table 4 details the Work Breakdown Structure for the Sentinel-IoT project, dividing the 25-week workload into six primary milestones. The methodology systematically progresses through two academic iterations. FYP 1 covers project initiation, the development of the AI engine [4], and XDR integration [1]. FYP 2 builds upon this foundation with the GRC dashboard development [13], rigorous system evaluation, and comprehensive documentation prior to the final defense.

10.2 Gantt Chart

The following Gantt chart visualizes the project timeline, illustrating the sequential nature of the development across the FYP 1 and FYP 2 semesters.

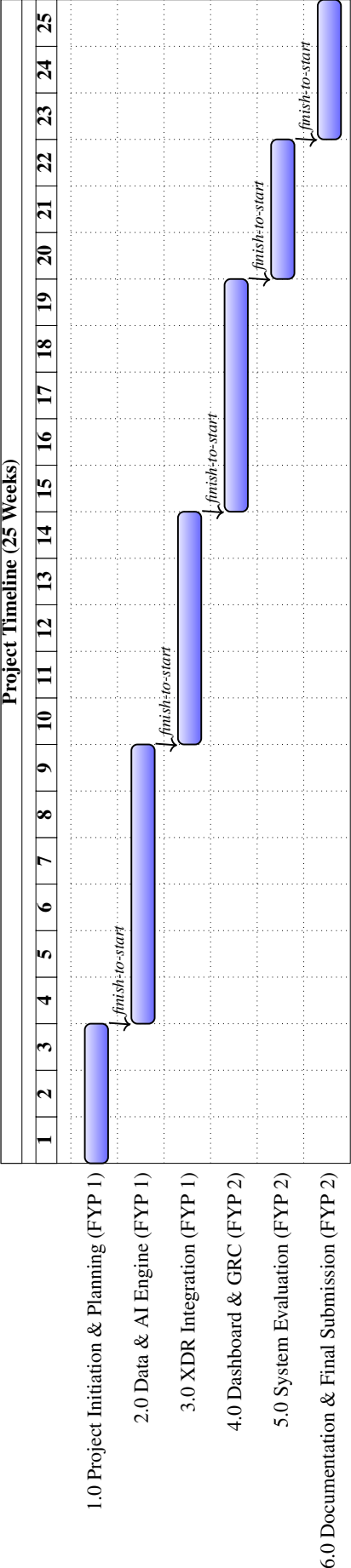


Figure 3: Project Gantt Chart for Sentinel-IoT

Figure 3 illustrates the 25-week development timeline for the Sentinel-IoT framework, structured across six sequential phases [2]. The project advances through two primary iterations: FYP 1 focuses on training the AI engine [4] and integrating Wazuh XDR for automated response [1]. FYP 2 concludes the framework by building the GRC dashboard [9], performing rigorous system evaluation, and finalizing documentation, ensuring full functionality before the academic defense [13].

11 Conclusion

The rapid expansion of Industrial IoT (IIoT) has introduced complex cyber-physical vulnerabilities that traditional, passive Intrusion Detection Systems are no longer equipped to handle [1]. To address this critical gap, this project proposes Sentinel-IoT, an autonomous Extended Detection and Response (XDR) framework [1]. By advancing the foundations of recent deep learning research, Sentinel-IoT transitions security from passive alerting to active, sub-second remediation [2, 3].

Integrating a highly accurate, hybrid AI engine (CNN-BiLSTM-Transformer) with the Wazuh XDR platform allows the system to not only detect zero-day anomalies but also autonomously enforce security policies, such as terminating malicious processes or blocking compromised IP addresses [11, 7]. Furthermore, by incorporating Explainable AI (SHAP) and a GRC-focused dashboard mapped to NIST and ISO standards, the framework ensures that automated decisions remain transparent, interpretable, and compliant for security analysts [14, 15].

Ultimately, successfully developing and evaluating this framework over the proposed 25-week timeline using the comprehensive ToN_IoT dataset will result in a robust, zero-trust security solution [12]. Sentinel-IoT will demonstrate how the fusion of advanced artificial intelligence and automated orchestration can effectively safeguard the next generation of critical industrial infrastructure [5, 13].

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